

Unit root tests

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Unit roots are nonstationary autoregressive (AR) or autoregressive moving average (ARMA) time series processes which may include an intercept and/or a trend. These processes are used often in economics and finance, but can also be found in other scientific fields. Unit root tests address the null hypothesis of a unit root, and an alternative hypothesis of a stationary (or trend stationary) time series. Critical values for unit root tests are typically derived via simulation of limiting distributions expressed as functionals of Brownian motions. The critical values for the Dickey Fuller unit root test with a constant and linear trend are derived via simulation in the R language. Simulation studies are presented showing that linear regressions with unit root processes often produce spurious results. Additional simulation studies are reviewed providing statistical evidence that near-unit roots can often result in spurious cointegration relationships. Various unit root tests are presented, including ones that allow for structural breaks in intercept and/or trend. Threshold unit root tests are introduced. Simulation studies are used to compare the unit root tests under various scenarios. The case where the analyzed time series may have stationary and nonstationary segments is also considered. © 2017 Wiley Periodicals, Inc.

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INTRODUCTION

Unit roots are nonstationary autoregressive (AR) or autoregressive moving-average (ARMA) time series processes; the simplest example is a random walk such as $x_t = x_{t-1} + \varepsilon_t$ where ε_t are random innovations. If we assume that the innovations are independent and identically distributed with $E(\varepsilon_t) = 0$ and $\text{var}(\varepsilon_t) = 1$, then $E(x_t) = x_0$ and $\text{var}(x_t) = t$; the process is nonstationary because the variance is a function of time. These processes are common in economics and finance. For example, as was originally proposed in the seminal work by Fam,¹ the logarithm of stock prices is often modeled as a random walk: $\log(S_t) = \log(S_{t-1}) + \varepsilon_t$ which is equivalent to modeling log returns as a stationary process: $\log\left(\frac{S_t}{S_{t-1}}\right) = \varepsilon_t$. Unit root processes can be extended to ARMA processes, where there are additional lagged innovation terms.

Unit root tests (URT) address the null hypothesis of a unit root; in the driftless AR(1) setting ($x_t = \phi_1 x_{t-1} + \varepsilon_t$) this statement is equivalent to assuming that $\phi_1 = 1$ under the null. The alternative hypothesis is often taken a stationary process with $|\phi_1| < 1$, however we will expand this to be $\phi_1 \neq 1$ which includes nonstationary processes referred to as explosive time series with $|\phi_1| > 1$. We will show using simulation studies that unit roots often result in spurious regressions. Additional simulation studies will show that near-unit roots where $\phi_1 = 1 + \delta$ or $\phi_1 = 1 - \delta$ for a small positive δ , will often lead to spurious cointegration relationships; this can happen when the URT used on the underlying time series fail to reject the null hypothesis of a unit root. Before we can cover URT in detail, we will discuss ARMA models, unit roots and cointegration.

Throughout this review continuous time models will not be considered, only discrete time models. However two continuous time processes related to unit roots that are commonly used in many fields would fail to reject the null hypothesis if a URT would be applied on them. The first such process is Brownian motion, also known as the Wiener process, which can be constructed

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as a re-scaled random walk. More specifically, a re-scaled sum of mean-zero, variance-one random variables ε_i as follows. Define $r \in [0, 1]$, the limit of the sum converges in distribution to a Wiener process:

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \sum_{i=1}^{\lfloor nr \rfloor} \varepsilon_i \Rightarrow W(r). \quad (1)$$

One of the properties of Brownian motion is that given $0 \leq t \leq s$, $W(s) - W(t)$ is independent of $W(u)$ where $u \leq t$ and has a Gaussian distribution of $N(0, s - t)$. So we can easily see that if we were to use a URT on Brownian motion it would fail to reject the null since we would be testing $W_t = \phi_1 W_{t-1} + \nu_t$ where ν_t is random and $\phi_1 = 1$. The other commonly used continuous time process is geometric Brownian motion, which does not allow negative values, assuming the initial value X_{t_0} is positive, where $t_0 = 0$:

$$X_t = X_0 e^{\mu t + \sigma W_t}, t \in [0, \infty). \quad (2)$$

If we consider a Taylor expansion of the exponential function $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$, then we can rewrite Equation (2) as:

$$X_t = X_0(1 + \mu t + \sigma W_t + \dots) = X_0 + X_0 f(W_t, \mu, \sigma, t). \quad (3)$$

Given that $f(W_t, \mu, \sigma, t)$ is a random function, it is very likely that if we perform a URT on x_t the null hypothesis of a unit root will fail to be rejected: $X_t = \phi_1 X_0 + \nu_t$ where ν_t is random and $\phi_1 = 1$.

STATIONARY TIME SERIES

A weakly stationary time series process is defined as having a constant mean, and an autocovariance function $\gamma(s, t)$ that depends on s and t only through their difference $|s - t|$. Throughout this document a stationary time series is intended to mean a weakly stationary time series. A strictly stationary time series is one where the joint probability distribution does not change with time. A strictly stationary process is rarely observed in practice. A nonstationary time series can have a time varying mean as well as a time varying covariance function.

A process that is integrated order n , $I(n)$, is a process that needs to be differenced n times to become weakly stationary. A weakly stationary process is referred to as an $I(0)$ process.

An ARMA(p,q) model is defined in terms of its lagged values $x(t)$ and its current and past innovations ε_t as:

$$x_t = \sum_{i=1}^p \phi_i x_{t-i} + \sum_{i=1}^q \theta_i \varepsilon_{t-i} + \varepsilon_t. \quad (4)$$

It is commonly assumed that the innovations are an independent identically distributed (i.i.d.) Gaussian white noise series $\varepsilon_t \sim N(0, \sigma^2)$, though this is not a requirement.

Equation (4) can be rewritten as:

$$x_t - \sum_{i=1}^p \phi_i x_{t-i} = \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i}, \quad (5)$$

and using the back-shift operator we can express Equation (5) as polynomial functions:

$$\phi(L)x_t = \theta(L)\varepsilon_t, \quad (6)$$

where $\phi(L)$ is the characteristic polynomial of the AR(p) part:

$$\phi(z) = 1 - \phi(1)z - \dots - \phi(p)z^p. \quad (7)$$

where x_t is stationary only if the roots of $\phi(L) = 0$ all lie outside of the unit circle.

Unit Roots

A unit root is defined as an AR or ARMA process that has 1 as a valid root of the characteristic polynomial equation as in Chan², p. 29. Time series with unit roots are nonstationary processes. In the case of an AR(1) process if $|\phi_1| = 1$ there will be a unit root, however the focus is usually on $\phi_1 = 1$; when $\phi_1 = -1$, even though the variance is not constant the process will exhibit an oscillatory behavior as the sign is reversed in every step, which we would argue is a less pathological case than when $\phi_1 = 1$. Consider the unit root $x_t = x_{t-1} + \varepsilon_t$; this is an integrated $I(1)$ process which can be turned into a stationary process via first differences: $\Delta(x_t) = \varepsilon_t$. As detailed in Chan,² if AR(p) process has all of its characteristic polynomial roots with an absolute value greater than one, then such as process is defined to be causal, and will also be stationary.

URT Hypotheses

URTs of a time series address the null hypothesis that the series is *unit root* nonstationary; that is, the

hypothesis the series is an integrated order 1 process, $I(1)$. The alternative hypothesis is that the time series is weakly stationary, or trend stationary in the case we consider a trend under the null. However another possibility is that the process becomes explosive when $|\phi_1| > 1$.

This article focuses on AR(1) models, with intercept and linear trend. The Augmented Dickey-Fuller (ADF) URT handles ARMA(p,q) models by rewriting them as an AR(1) model with additional components of differences; after essentially ‘taking out’ these additional components in a helper regression it is then able to use the critical values of the Dickey-Fuller that were designed with an AR(1) model.

Null URT Hypothesis

Unless otherwise noted, the null hypothesis is defined as an AR(1) model $x_t = \phi_1 x_{t-1} + \varepsilon_t$ where $\phi_1 = 1$. This corresponds to a nonstationary $I(1)$ process. This is unusual in statistics where we would typically consider the multiplier to be 0 under the null as in a t-test for the coefficient of a linear regression model. The reason $\phi_1 = 1$ is picked as the null is because many time series in finance and economics are unit roots.

The case where the null hypothesis in the AR(1) model is extended to allow an intercept and a linear trend as follows will also be considered, in a one-dimensional state-space model like representation:

$$\begin{aligned} x_t &= \phi_1 x_{t-1} + \varepsilon_t, \\ y_t &= \beta_0 + \beta_1 t + x_t, \\ \phi_1 &= 1. \end{aligned} \quad (8)$$

Enhancing the URT null hypothesis to allow for structural breaks/changepoints in the deterministic coefficients β_0 and β_1 will also be discussed.

If, instead of considering the intercept and trend in a state-space model framework as before, we had allowed the trend in the AR formula under the null, this would result in quadratic time trends in the levels. Consider:

$$\begin{aligned} x_t &= \phi_1 x_{t-1} + \beta_0 + \beta_1 t + \varepsilon_t, \\ \phi_1 &= 1. \end{aligned} \quad (9)$$

Assuming that $x_t = 0$, if we solve the recursion, using Faulhaber’s formula we would find the following expression in the levels of $x(t)$ with a quadratic time trend component:

$$x_t = \beta_0 t + \beta_1 \frac{t(t+1)}{2} + \sum_{i=1}^t \varepsilon_i. \quad (10)$$

Alternative URT Hypothesis

Unless otherwise specified, the alternative hypothesis is assumed to be AR(1) with $\phi_1 \neq 1$. When $|\phi_1| < 1$ this corresponds to a stationary $I(0)$ process. Most of the standard URTs, such as the ADF test consider the alternative hypothesis to be $|\phi_1| < 1$. However in the case with an explosive AR(1) process with $|\phi_1| > 1$ this is a highly nonstationary process. In this article we will only consider the case when $\phi_1 > 1$. Testing for cointegration with AR(1) explosive series can quite easily lead to a spurious cointegration result as will be shown using a simulation study later. This is why it is beneficial to have URTs with power against the explosive alternative hypothesis.

When allowing an intercept and a linear trend in the time series being tested for unit roots the same one-dimensional state-space model like representation as under the null is assumed:

$$\begin{aligned} x_t &= \phi_1 x_{t-1} + \varepsilon_t, \\ y_t &= \beta_0 + \beta_1 t + x_t, \\ \phi_1 &\neq 1. \end{aligned} \quad (11)$$

If we allowed a drift and a trend in the AR formula under the alternative ($x_t = \phi_1 x_{t-1} + \beta_0 + \beta_1 t + \varepsilon_t$), assuming that $x_0 = 0$, the resulting form would be $y_t = \beta_0 + \beta_1 t + \sum_{i=1}^t \phi_1^{t-i} \varepsilon_i$, which will be stationary if $|\phi_1| < 1$ and explosive if $|\phi_1| > 1$.

The case where the URT alternative hypothesis allows for structural breaks/changepoints in the deterministic coefficients β_0 and β_1 will be reviewed later in this article.

Cointegration

Linear relationships involving integrated nonstationary time series are meaningful only if the time series are cointegrated. There are various definitions of cointegration; as originally proposed by Engle and Granger,³ a cointegrating relationship between two or more time series each having unit roots ($I(1)$ $I(1)$) is defined to exist if there is a linear combination that is stationary, i.e., $I(0)$. Two time series x_t and y_t are cointegrated when there exists a number α_1 in the linear equation $y_t = \alpha_1 x_t + \nu_t$ such that ν_t is a stationary process. This approach can be implemented in the two variable case using the two-step Engle Granger procedure, where the first step consists of using least squares to derive a linear relationship between the two variables, and the second step consists of using a URT on the residuals of the first step’s regression. When there are more than two time series to check for cointegration, there are multiple possible

cointegrating relationships, and the Engle Granger two step methodology is not sufficiently flexible. In this case the most common approach is to use the Johansen cointegration test, which is based on the estimation of a p th-order VAR in the k variables. The VAR in the k -vector y is:

$$y_t = \Pi_1 y_{t-1} + \Pi_2 y_{t-2} + \dots + \Pi_p y_{t-p} + \Psi D_t + \varepsilon_t. \quad (12)$$

where D_t is a d -vector of deterministic terms, such as a constant, time trend and seasonal dummies if necessary. The estimation of the rank of the following matrix Π will approximate the number of possible cointegrating relationships if any as follows:

$$\Pi = -(I - \Pi_1 - \Pi_2 - \dots - \Pi_p). \quad (13)$$

No assumption is made about the rank of Π . In the decomposition $\Pi = \alpha\beta'$, α and β are $k \times k$ matrices. The goal is to determine whether any columns of β' are statistically indistinguishable from zero vectors. The existence of r cointegrating vectors reduces the rank of Π by $k - r$: that is, if there were r cointegrating relationships between the given variables, then there would be r nonzero eigenvalues in the dynamic system, and $k - r$ zero eigenvalues. See Johansen, S. and Juselius, K.⁴ for details. The methodology is based on canonical correlation analysis.

Threshold Cointegration

Consider a general cointegration framework of two time series y_t and x_t :

$$y_t + \alpha x_t = z_t. \quad (14)$$

Balke and Fomby⁵ proposed a general equilibrium model of z_t based on a self exciting threshold autoregressive (SETAR) framework such as the following with a low, middle and high regimes:

$$z_t = \begin{cases} \mu_h + \phi_h z_{t-1} + \varepsilon_t, & \text{if } z_{t-1} > \theta_H, \\ \mu_m + \phi_m z_{t-1} + \varepsilon_t, & \text{if } \theta_L \leq z_{t-1} \leq \theta_H, \\ \mu_l + \phi_l z_{t-1} + \varepsilon_t, & \text{if } z_{t-1} < \theta_L. \end{cases} \quad (15)$$

As Stigler⁶ points out the common assumed case where Equation (15) is stable is when $\phi_h < 1$ and $\phi_l < 1$.

Seo⁷ developed a test for the linear no cointegration null hypothesis against threshold cointegration in a threshold vector error correction model. They used a sup-Wald type test and derived its null asymptotic distribution. They proposed using a

residual-based bootstrap and showed with Monte Carlo simulations that the bootstrap corrected size distortions of the asymptotic distribution in finite samples, and that it had greater power against the threshold cointegration alternative than conventional cointegration tests.

Stigler⁶ posits that SETAR models are so complex that their probability distribution and moments are only known for simple cases. Estimation for more than one threshold is not easy. He points out that in a cointegration model with threshold effects there is no known procedure to test for stationarity with an unknown cointegrating vector. A test to determine 2 versus 3 regimes only works in a restricted case.

Bec et al.⁸ proposed the BBC URT which considers a three-regime SETAR model. They allow a unit root in the middle regime in the middle regime under the alternative hypothesis, but not for the high and low regimes. Bec et al. used the sup-Wald, sup-LM and sup-LR statistics to derive critical values.

NOTATION AND ABBREVIATIONS

All of the simulations presented in this article were developed using the R language. The R language as well as many extension packages can be downloaded from <https://cran.r-project.org/> for various computer architectures including Windows, MacOS and Linux. The following symbols and abbreviations are used:

- The number of time steps in a time series is denoted by l
- The number of simulations is indicated by m
- The significance level of the statistical test is designated by α ; it will be typically 0.05
- The seed used for random number generation in this article is 12345 for each simulation, unless otherwise noted
- Each simulation follows a specified data generation process (DGP)
- $\chi \sim N(0, 1)$: A Gaussian distributed random variable with mean 0 and variance 1
- $\chi \sim U(a, b)$: A uniformly distributed random variable between a and b where $a < b$

IMPORTANCE OF URTS

URTs are used to determine if time series are likely nonstationary or not; more specifically in the case of

an AR(1) model, most test the null $\phi_1 = 1$ in $x_t = \phi_1 x_{t-1} + \varepsilon_t$, and the alternative hypothesis is typically $|\phi_1| < 1$. However as we will see it is useful to consider the alternative hypothesis to be $\phi_1 \neq 1$ since not only when $\phi_1 < 1$ do spurious regressions occur, but they also can happen when $\phi_1 > 1$. We will also see that when $\phi_1 > 1$ many URTs will not reject the null of a unit root, and almost invariably a cointegration test of independent variables with $\phi_1 > 1$ would result in spurious cointegration relationships.

Spurious Regressions

Consider the following trivial DGP of linearly independent x_t and y_t series:

$$\begin{aligned} x_t &= gx_{t-1} + \varepsilon_t, \\ y_t &= hy_{t-1} + \gamma_t, \\ x_0 &= y_0 = 0, \\ \varepsilon_t &\sim i.i.d. \sim N(0,1), \gamma_t \sim i.i.d. \sim N(0,1). \end{aligned} \quad (16)$$

Using the results of the DGP (16) the regression $Y_t = \beta X_t + \nu_t$ is performed in a simulation with $m = 1000$ replications. Table 1 summarizes the regression results, and we see for the median (0.5 quantile) that in the cases of $g = 1.01$, $h = 1.01$ and $g = 1$, $h = 1$, the regressions appear to be significant even though Y_t and X_t are independent of each other. Even in the case of $g = 0.9$, $h = 0.9$ with an F -stat p -value of 0.04 leads us to reject the null hypothesis of $\beta = 0$ at the 0.05 significance level. Granger⁹, p. 13, confirms similar findings: that spurious regressions can happen for non $I(1)$ processes (with an AR(1) ϕ_1 multiplier of 0.9.)

TABLE 1 | Regressing y_t on x_t as Defined in Equation (16) with $l = 1000$ and $m = 1000$

g	h	Adj R^2 Q:0.5	Fstat p -value Q:0.5
1.01	1.01	1.0000	0.00
1.00	1.00	0.4045	0.00
0.90	0.90	0.034	0.04
0.00	1.00	-0.006	0.50
0.50	1.00	0.003	0.25
0.50	0.50	-0.002	0.38
0.20	0.20	-0.005	0.47
0.00	0.00	-0.006	0.50

Spurious Cointegration

$$\begin{aligned} x_t &= \phi_1^a x_{t-1} + \varepsilon_t, \\ y_t &= \phi_1^b y_{t-1} + \zeta_t, \\ x_0 &= y_0 = 0, \\ \varepsilon_t &\sim i.i.d. \sim N(0,1), \\ \zeta_t &\sim i.i.d. \sim N(0,1). \end{aligned} \quad (17)$$

We use DGP (17), where x_t and y_t are independent, to perform simulation tests of spurious cointegration where the Johansen cointegration test is performed on the two series, without considering if they are unit roots or not.

We can see that in Table 2 for the case of $\phi_1^a = 1$ and $\phi_1^b = 1$ the failure to reject the null hypothesis of no-cointegration is close to the significance level. However for the case of $\phi_1^a > 1$ and/or $\phi_1^b > 1$ the failure to reject the null hypothesis of no-cointegration are quite high—providing evidence of cointegration in 90–100% of the cases which is entirely spurious and invalid since a valid cointegration relationship requires that the tested variables are unit root processes. The same is true if $\phi_1^a < 0.9$ and/or $\phi_1^b < 0.9$ but in these cases most of the URT's would likely reject the null hypothesis of a unit root, so these cases would not be as troublesome as when $\phi_1^a > 1$ and/or $\phi_1^b > 1$ where the standard URT's such as ADF would not reject the null hypothesis of a unit root.

URT SURVEY

The following is a survey of some of the most commonly used URTs, as well as some others. This is not an exhaustive review, as there are too many URTs to cover in a single article. For more thorough reviews see Refs 10–12.

Dickey Fuller URT

The original simplest version of the Dickey Fuller test considered the standard AR(1) model with Gaussian white noise without any deterministic components; see Ref 13 for details:

$$\begin{aligned} y_t &= \phi_1 y_{t-1} + \varepsilon_t; \\ \varepsilon_t &\sim i.i.d. \sim N(0,1); y_0 = 0. \end{aligned} \quad (18)$$

As detailed in Patterson¹⁴, p. 208, the two main test statistics for the Dickey Fuller test are the normalized bias $\hat{\delta}$ and the t-type statistic $\hat{\tau}$:

TABLE 2 | Proportions of Failures to Reject the Null Hypothesis of No Cointegration of Pairs of AR(1) Time Series with $m = 5000$, $I = 1000$ and $\alpha = 0.05$ as Defined in Model (17)

ϕ_1^a	ϕ_1^b	Johansen $r = 0$
1.010	1.010	1.00
1.000	1.020	1.00
1.000	1.010	1.00
1.000	1.005	0.94
1.000	1.000	0.05
0.990	0.990	0.17
1.000	0.960	0.97
1.000	0.970	0.80
1.000	0.980	0.42
1.000	0.990	0.12
1.000	0.900	1.00
1.000	0.200	1.00
0.900	0.900	1.00
0.500	0.500	1.00
0.200	0.200	1.00

$$\hat{\delta} = T(\hat{\phi}_1 - 1), \quad (19)$$

$$\hat{\tau} = \frac{(\hat{\phi}_1 - 1)}{\hat{\sigma}_{\hat{\phi}}}, \quad (20)$$

where $\hat{\phi}_1$ is the standard coefficient estimate as derived using ordinary least squares:

$$\hat{\phi}_1 = \frac{\sum_{t=1}^T y_t y_{t-1}}{\sum_{t=1}^T y_{t-1}^2}, \quad (21)$$

$$\hat{\phi}_1 - 1 = \frac{\sum_{t=1}^T y_{t-1}(y_t - y_{t-1})}{\sum_{t=1}^T y_{t-1}^2} = \frac{\sum_{t=1}^T y_{t-1} \varepsilon_t}{\sum_{t=1}^T y_{t-1}^2}. \quad (22)$$

The last step in Equation (22) is based on $\varepsilon_t = y_t - y_{t-1}$.

Given these previous equations we can now define the normalized bias as a function of the observations y_t :

$$\hat{\delta} = T \frac{\sum_{t=1}^T y_{t-1} \varepsilon_t}{\sum_{t=1}^T y_{t-1}^2} = \frac{\sum_{t=1}^T y_{t-1} \varepsilon_t / T}{\sum_{t=1}^T y_{t-1}^2 / T^2}. \quad (23)$$

As detailed in Table 7.1 of Ref 14, p. 197 for the DGP (18) following sample quantities have a limiting form expressed as functionals of Brownian motion:

$$\sum_{t=1}^T y_{t-1} \varepsilon_t / T \xrightarrow{D} \sigma^2 \int_0^1 W_t dW_t, \quad (24)$$

$$\sum_{t=1}^T y_{t-1}^2 / T^2 \xrightarrow{D} \sigma^2 \int_0^1 W_t^2 dr. \quad (25)$$

These expressions can be simulated to determine critical values for these test statistics.

Most standard URTs in the AR(1) framework, such as the original Dickey-Fuller test, have a null hypothesis of $\phi_1 = 1$ and the alternative hypothesis is $|\phi_1| < 1$. The case of a negative unit root $\phi_1 = -1$ leads to a distribution of $(\hat{\phi}_1 + 1)$ that is the mirror image of the standard Dickey-Fuller distribution as detailed in Ref 10, p. 55. The situation when $|\phi_1| > 1$ this leads to an explosive case. As derived in Phillips and Magdalinos¹⁵ the asymptotics of the estimation error of ϕ_1 in this case under certain assumptions is Cauchy distributed. This distribution is problematic as it has no finite moments of order greater than or equal to one.

We can extend the DGP (18) to have a constant and a linear trend; Using OLS these extra terms can be removed. This will affect the critical values of the test statistics. Zivot¹⁶ suggests a simulation approach based in R. This approach was extended for the trend and drift terms for the normalized bias (NB) and t-type (DF) test statistics; the R code to perform these simulations is detailed in Listing 1.

Results of running the simulation code from Listing 1 and comparisons with the critical values from the R `qunitroot()` function from the `fUnitRoots` R package as reported by MacKinnon¹⁷ are detailed in Listing 2:

We see in Listing 2 that the simulated values and the ones computed by MacKinnon¹⁷ as reported in the `qunitroot()` function of the `fUnitRoots` R package are relatively close.

Said and Dickey¹⁸ extended the Dickey Fuller URT for ARMA models and not just AR(1) models; this is known as the Augmented Dickey Fuller (ADF) URTs and is one of the most commonly used URTs in the literature Ref 10, p. 33. The ADF test regression is fitted using OLS:

$$\Delta y_t = \alpha + \delta t + \beta y_{t-1} + \sum_{i=1}^n \gamma_i \Delta y_{t-i} + \varepsilon_t, \quad (26)$$

where Δ is the difference operator and ε_t represent 0-mean white-noise innovations. Under the null

Listing 1

NB and DF Test Statistic Simulations

```

library(fUnitRoots); library(dynlm); library(zoo)
wiener_bar <- function(nobs) {
  e <- rnorm(nobs); drift <- runif(1,-20,20)
  #Model is  $y_t = C + D * t + \phi_{t-1} * y_{t-1} + \epsilon_t$ 
  #HO: we assume that  $\phi_{t-1} = 1$  and  $D=0$ 
  y <- cumsum(e) + cumsum( rep(drift,nobs) )
  tsy <- zoo(y); l <- length( tsy[,1] )
  ul <- 1:l; model1 <- dynlm(formula = tsy~ ul )
  D <- model1$coefficients[ 'ul' ]
  C <- model1$coefficients[ '(Intercept)' ]
  #demean and detrend
  y <- y - rep(C,nobs) - D*ul
  yml <- y[1:(nobs-1)]
  intW2 <- nobs^(-2) * sum(yml^2)
  intWdW <- nobs^(-1) * sum(yml*e[2:nobs])
  ans <- list(intW2=intW2,intWdW=intWdW); ans
}
set.seed(12345); nobs <- 1000; nsim <- 10000
NB <- rep(0,nsim); DF <- rep(0,nsim)
for (i in 1:nsim) {
  BN.moments <- wiener_bar(nobs)
  NB[i] <- BN.moments$intWdW/BN.moments$intW2
  DF[i] <- BN.moments$intWdW/sqrt(BN.moments$intW2)
}
quantile(DF,probs=c(0.01,0.05,0.1))
qunitroot(c(0.01,0.05,0.10), trend='ct', statistic='t', N=10000)
quantile(NB,probs=c(0.01,0.05,0.1))
qunitroot(c(0.01,0.05,0.10), trend='ct', statistic='n', N=10000)

```

hypothesis y_t is considered to be $I(1)$ which is equivalent to Δy_t being $I(0)$ in which case β would be zero. The test statistic is the standard regression t -statistic $t_\beta = \frac{\hat{\beta}}{s.e.(\hat{\beta})}$. A normalized bias test statistic can be used as well.

The tests' authors derived the critical values for these test statistics as they follow a nonstandard distribution; they are the same as the standard Dickey-Fuller critical values which depend on the form of the deterministic components. The lagged differences allow correcting for serial correlation in the innovations. The ADF URT is sensitive to the choice of the number of lags n . Ng and Perron¹⁹ proposed an iterative method to determine the lag length based on choosing the largest lag where the

t -ratio of γ_i is still significant. For the ADF test details see Ref 18.

Elliott, Rothenberg and Stock URTs

See Ref 20, p. 93, for a detailed overview of the Elliott, Rothenberg and Stock (ERS) URTs; the following overview is taken from there. To increase the power of a URT under the null hypothesis of a unit root, ERS [1996] proposed a local to unity detrending of the time series. The assumed generating process for the series y_t is as follows:

$$y_t = d_t + u_t \quad u_t = \alpha u_{t-1} + v_t \quad d_t = \hat{\beta}' z_t, \quad (27)$$

Listing 2

Critical Values of Simulated DF Statistics

```

> quantile(DF, probs=c(0.01, 0.05, 0.1))
      1%      5%     10%
-3.980987 -3.406626 -3.125696
> qunitroot(c(0.01, 0.05, 0.10), trend='ct',
  statistic='t', N=10000)
[1] -3.958799 -3.410295 -3.126858
> quantile(NB, probs=c(0.01, 0.05, 0.1))
      1%      5%     10%
-29.29784 -21.59540 -18.13030
> qunitroot(c(0.01, 0.05, 0.10), trend="ct",
  statistic="n", N=10000)
[1] -29.32316 -21.68862 -18.23051

```

where z_t is a deterministic $(q \times 1)$ vector and ν_t is a stationary zero-mean process. If $a = 1$ the y_t is $I(1)$, but if $|a| < 1$ then y_t is $I(0)$.

The authors developed feasible point-optimal tests, which take serial correlation of the error term into account. The feasible point-optimal test statistic is defined as:

$$P_T = \frac{S(a = \bar{a}) - \bar{a}S(a = 1)}{\hat{\omega}^2}, \quad (28)$$

where $S(a = \bar{a})$ and $S(a = 1)$ are the sums of squared residuals from a least-squares regression of y_a on Z_a with $y_a = (y_1, y_2 - ay_1, \dots, y_T - ay_{T-1})$ and $Z_a = (z_1, z_2 - az_1, \dots, z_T - az_{T-1})$ and y_a is a T -dimensional vector and Z_a is a $(T \times q)$ matrix. The estimator for the variance of the error process ν_t is:

$$\hat{\omega} = \frac{\hat{\sigma}_\nu^2}{\left(1 - \sum_{i=1}^p \hat{\alpha}_i\right)^2}, \quad (29)$$

where $\hat{\sigma}_\nu^2$ and $\hat{\alpha}_i$ for $i = 1, \dots, p$ are from the OLS regression:

$$\Delta y_t = \alpha_0 + \alpha_1 \Delta y_{t-1} + \dots + \alpha_{p+1} \Delta y_{t-p} + \nu_t \quad (30)$$

And $\bar{a} = 1 + \bar{c}/T$ where $\bar{c}$ is a constant, set to -7 in the case of a constant or -13.5 in the case of a linear trend. The second test type is denoted as the DF-GLS test, which is a modified ADF-type test applied to the detrended data without the intercept, which is the t -statistic for testing $\alpha_0 = 0$ in the regression:

$$\Delta y_t^d = \alpha_0 y_{t-1}^d + \alpha_1 \Delta y_{t-1}^d + \dots + \alpha_p \Delta y_{t-p}^d + \varepsilon_t, \quad (31)$$

where y_t^d are residuals from $y_t - \hat{\beta}' z_t$

Both of these tests are available in the `urca` package in the `ur.ers()` function. Later in this article we will see in simulation experiments that these ERS tests become invalid when the initial value y_0 is not 0, and also when the $AR(1)$ ϕ_1 multipliers are negative.

Phillips-Perron URT

Phillips and Perron²¹ developed a URT that corrects for serial correlation and heteroskedasticity of the innovations. Pfaff²⁰, p. 95, reviews the Phillips-Perron (PP) procedure details. The PP test regression is fitted using OLS. The first case is with out a linear-trend:

$$y_t = \mu + \alpha y_{t-1} + \varepsilon_t, \quad (32)$$

$$Z(\hat{\alpha}) = T(\hat{\alpha} - 1) - \frac{\hat{\lambda}}{\bar{m}_{yy}}, \quad (33)$$

$$Z(\tau_{\hat{\alpha}}) = \frac{\hat{s}t_{\hat{\alpha}}}{\hat{\sigma}_{TI}} - \frac{\hat{\lambda}' \hat{\sigma}_{TI}}{\bar{m}_{yy}^{\frac{1}{2}}}, \quad (34)$$

$$Z(\tau_{\hat{\mu}}) = \frac{\hat{s}t_{\hat{\mu}}}{\hat{\sigma}_{TI}} - \frac{\hat{\lambda}' \hat{\sigma}_{TI} m_y}{\bar{m}_{yy}^{\frac{1}{2}} m_y^{\frac{1}{2}}}, \quad (35)$$

with $\bar{m}_{yy} = T^{-2} \sum (y_t - \bar{y})^2$, $m_{yy} = T^{-2} \sum (y_t^2)$, $m_y = T^{-3/2} \sum (y_t)$, and $\hat{\lambda} = 0.5(\hat{\sigma}_{TI}^2 - \hat{s}^2)$, where $\hat{s}^2$ is the sample variance of the residuals, $\hat{\lambda}' = \hat{\lambda}/\hat{\sigma}_{TI}^2$, and $t_{\hat{\alpha}}$, $t_{\hat{\mu}}$ are the student t ratios of $\hat{\alpha}$, $\hat{\mu}$, respectively. The long run variance $\hat{\sigma}_{TI}^2$ is estimated as follows:

$$\hat{\sigma}_{TI}^2 = T^{-1} \sum_{t=1}^T \hat{\varepsilon}_t^2 + 2T^{-1} \sum_{s=1}^l w_{sl} \sum_{t=s+1}^T \hat{\varepsilon}_t \hat{\varepsilon}_{t-s}, \quad (36)$$

where $w_{sl} = 1 - s/(l+1)$.

The case of adding a linear trend is now considered:

$$y_t = \mu + \beta(t - \frac{T}{2}) + \alpha y_{t-1} + \varepsilon_t. \quad (37)$$

In this case the test statistics used are:

$$Z(\hat{\alpha}) = T(\hat{\alpha} - 1) - \frac{\hat{\lambda}}{M}, \quad (38)$$

$$Z(\tau_{\hat{\alpha}}) = \frac{\hat{s}t_{\hat{\alpha}}}{\hat{\sigma}_{TI}} - \frac{\hat{\lambda}' \hat{\sigma}_{TI}}{M^{\frac{1}{2}}}, \quad (39)$$

$$Z(\tau_{\hat{\mu}}) = \frac{\hat{s}t_{\hat{\mu}}}{\hat{\sigma}_{TI}} - \frac{\hat{\lambda}' \hat{\sigma}_{TI} m_y}{M^{\frac{1}{2}} (M + m_y^2)^{\frac{1}{2}}}, \quad (40)$$

$$Z(\tau_{\hat{\beta}}) = \frac{\hat{s}t_{\hat{\beta}}}{\hat{\sigma}_{TI}} - \frac{\hat{\lambda}' \hat{\sigma}_{TI} (0.5m_y - m_{ty})}{(\frac{M}{12})^{\frac{1}{2}} m_{yy}^{\frac{1}{2}}}, \quad (41)$$

where m_y , $\bar{m}_{yy}$, $\hat{\lambda}$, $\hat{\lambda}'$ and $\hat{\sigma}_{TI}$ are as defined previously and $m_{ty} = T^{-5/2} \sum t y_t$ and $t_{\hat{\mu}}$, $t_{\hat{\beta}}$, $t_{\hat{\alpha}}$ are the student t ratios of $\hat{\mu}$, $\hat{\beta}$, $\hat{\alpha}$ respectively. The scalar M is defined as $M = (1 - T^{-2})m_{yy} - 12m_{ty}^2 + 12(1 + T^{-1})m_{ty}m_y - (4 + 6T^{-1} + 2T^{-2})m_y^2$. The critical values used are the same as for the Dickey Fuller URT.

Lagged-Series URT

Herranz²² proposed a new URT for AR(1) time series referred to as the lagged-series test which checks for the number of cointegrating relationships

of a time series with the lagged time series using the Johansen cointegration test. *If the time-series is a unit root, $I(1)$, we expect only one valid cointegrating relationship but if there are two cointegrating relationships the process is $I(0)$.* Given two variables, only one valid cointegration relationship is possible. For reasonable lags it would be almost impossible for the two time series to fail to reject the null hypothesis of no-cointegration since the series and its lagged version will always be fairly close to each other. The lagged-series URT is conducted as follows:

- Decide if a constant and/or linear trend is allowed under the null hypothesis or not.
- In the case a constant and/or linear trend is allowed under the null hypothesis follow these steps:
 - Use a helper regression to estimate and remove these deterministic elements α and β in $x_t = y_t + \alpha + \beta t$ where the underlying AR process is defined as $y_t = \phi_1 y_{t-1} + \varepsilon_t$. The helper regression to be performed is:

$$\Delta x_t = \phi_1^a x_{t-1} + C + Dt + \varepsilon_t. \quad (42)$$

- The coefficients for the nondifferenced relationship can be derived as

$$\begin{aligned} \phi_1 &= \phi_1^a + 1; \beta = D/(1 - \phi_1); \alpha \\ &= (C - \beta \phi_1)/(1 - \phi_1). \end{aligned} \quad (43)$$

- Remove the deterministic elements as follows:

$$x_t = x_t - (\alpha + \beta) - \beta t. \quad (44)$$

- In the case where no constant and no trend is allowed under the null hypothesis the underlying AR process to be tested is defined as $x_t = \phi_1 x_{t-1} + \varepsilon_t$
- Run the Johansen cointegration test of the time series $\{x_t^*\} = \{x_{k+1}, \dots, x_n\}$ and the k-lagged time series $\{L^k(x_t)\} = \{x_1, x_2, \dots, x_{n-k}\}$ for a small lag, such as $k = 3$.
- The Johansen test computes a primary and a secondary test statistic when run with two variables; the first test statistic is for the null hypothesis of a cointegration rank of 0. The second test statistic is for a null-hypothesis of a cointegration rank of less than or equal to 1. Examine the *second* cointegration test

statistic of a rank of less than or equal to 1. If this test statistic is less than or equal to the critical value for the specified significance level α then do not reject the null of $I(1)$. Otherwise, if the statistic is greater than the critical value reject the null of $I(1)$ and accept the alternative of $I(0)$ or an explosive process ($|\phi_1| > 1$).

- The Johansen cointegration test has different critical values depending on whether the estimated relationship is assumed to have a constant, a linear trend or is driftless. We use the Johansen critical values corresponding to the driftless case when there are no deterministic elements. If we allowed for a constant and/or a trend then the Johansen critical values corresponding to the trend case multiplied by 1.04 are used. These critical values were determined via various simulation studies, under various scenarios and time series lengths.

The Zivot Andrews (ZA) URT, or many other URTs such as ADF, can be combined with the lagged-series URT as detailed in Algorithm 1 to increase the statistical power of the ZA test in the case where AR(1) process is explosive, that is $|\phi_1| > 1$. The deterministic component can incorporate either an intercept, time trend, or both. The value of C was determined via simulation studies.

Algorithm 1 ZA-lagged-series URT

```

C ← 20.09299
x0 ← LaggedSeriesUnitRootTest( tsx, lag, cvalLevel)@Statistic
x1 ← ur.za(tsx,model='trend')@Statistic
if x1 > CriticalValue and x0 < C then
    return( 1(1) ) /* Unit Root Nonstationary */
else
    return( 1(0) or Explosive) /* Stationary or  $|\phi_1| > 1$  */
end if

```

Kwiatkowski Phillips Schmidt Shin (KPSS) Stationarity Test

The KPSS test proposed by Ref 23, has the null hypothesis of stationarity around a deterministic trend and the alternative of a unit root. The following definitions and explanations of the KPSS test are taken from Ref 20, p. 103. The underlying model considered in the KPSS test is:

$$y_t = \xi t + r_t + \varepsilon_t, \quad (45)$$

$$r_t = r_{t-1} + u_t, \quad (46)$$

where r_t is a random walk and the innovations are assumed to be i.i.d. $(0, \sigma_u^2)$. r_0 is assumed to be a constant level. If $\xi < 0$ then the model will contain a time trend component in addition to the level. Under the null hypothesis y_t is assumed to be $I(0)$ so that $\sigma_u^2 = 0$ and $r_t = r_0 = \text{constant}$.

The test statistic is derived as follows. First y_t is regressed on a constant or a constant plus a linear trend. The partial sums S_t of the residuals $\hat{\varepsilon}_t$ are computed:

$$S_t = \sum_{i=1}^t \hat{\varepsilon}_i, t = 1, \dots, T. \quad (47)$$

The test statistic LM is computed as:

$$LM = \frac{\sum_{t=1}^T S_t^2}{\hat{\sigma}_\varepsilon^2}, \quad (48)$$

where $\hat{\sigma}_\varepsilon^2$ is an estimate of the error variance of the residuals. The authors suggest using a Bartlett window $w(s, l) = 1 - s/(l + 1)$ as an optimal weighting function for estimating the long run variance of the errors:

$$\hat{\sigma}_\varepsilon^2 = T^{-1} \sum_{t=1}^T \hat{\varepsilon}_t^2 + 2T^{-1} \sum_{s=1}^l \frac{s}{l+1} \sum_{t=s+1}^T \hat{\varepsilon}_t \hat{\varepsilon}_{t-1}. \quad (49)$$

The critical values of the test statistic are given in Ref 23.

Zivot Andrews URT

Andrews and Zivot²⁴ developed a URT (ZA) that can handle structural breaks. The test statistic of the ZA test is the Student t ratio just as was the case with the PP test. As detailed in Ref 20, p. 110:

$$t_{\hat{\alpha}i} \left[\hat{\lambda}_{inf}^i \right] = \inf_{\lambda \in \Delta} t_{\hat{\alpha}i}(\lambda) \text{ for } i = A, B, C, \quad (50)$$

where Δ is a closed subset of $(0, 1)$. Depending on the model, the test statistic is inferred from one of these three regression Models:

$$y_t = \hat{\mu}^A + \hat{\theta}^A D U_t(\hat{\lambda}) + \hat{\beta}^A t + \hat{\alpha}^A t y_{t-1} + \sum_{i=1}^k \hat{c}_i^A \Delta y_{t-i} + \hat{\varepsilon}_t, \quad (51)$$

$$y_t = \hat{\mu}^B + \hat{\gamma}^B D T_t^*(\hat{\lambda}) + \hat{\beta}^B t + \hat{\alpha}^B t y_{t-1} + \sum_{i=1}^k \hat{c}_i^B \Delta y_{t-i} + \hat{\varepsilon}_t, \quad (52)$$

$$y_t = \hat{\mu}^C + \hat{\theta}^C DU_t(\hat{\lambda}) + \hat{\beta}^C t + \hat{\alpha}^C t y_{t-1} + \hat{\gamma}^B DT_t^*(\hat{\lambda}) + \sum_{i=1}^k \hat{c}_i^C \Delta y_{t-i} + \hat{\varepsilon}_t, \quad (53)$$

where $DU_t(\lambda) = 1$ if $t > T\lambda$ and 0 otherwise, and $DT_t^*(\lambda) = t - \lambda T$ for $t > T\lambda$ and 0 otherwise. The null hypothesis of the ZA URT does not allow structural breaks. Only breaks are allowed in the alternative hypothesis. Glynn et al.²⁵ criticize this since if there are breaks under the null ($\phi_1 = 1$) we can mistakenly conclude that the series is stationary (with breaks). The critical values are derived via simulation of the t -test statistic under the null, and in the asymptotic case can be derived as a functional of Brownian motions.

Removing Structural Changes Prior to Unit Root Testing

Estimating Change Points

The Bai and Perron²⁶ procedure for estimating structural breaks/change points in a linear model is implemented in the *breakpoints()* function of the *strucchange* R package (Ref 27, p. 12).

This methodology assumes an underlying linear model with a dependent one dimensional variable y_t , a $p \times 1$ covariates vector $\mathbf{x}_t$ with corresponding coefficient vector $\boldsymbol{\beta}$, and the innovations ε_t :

$$y_t = \mathbf{x}_t \boldsymbol{\beta} + \varepsilon_t. \quad (54)$$

If there are m change points in the coefficient this implies $m + 1$ regimes. Equation (54) can be rewritten as:

$$y_t = \mathbf{x}_t \boldsymbol{\beta}_j + \varepsilon_t \quad (j = T_{j-1} + 1, \dots, T_j, j = 1, \dots, m + 1). \quad (55)$$

The underlying idea is solving the problem by dividing it into independent optimally solvable sub-problems, whose solutions can be combined to solve the larger problem. In this case a triangular residual sum of squares (RSS) matrix is computed and stored in memory which can be reused over and over again to derive the RSS for a segment starting at observation t and ending at t' with $t < t'$.

This approach is considerably faster than the brute force approach of computing the RSS for all possible sub-segments. The Bai and Perron²⁶ algorithm uses only $O(T^2)$ least squares operations for a number m of change points. The brute force approach would require $O(T^m)$ least squares operations.

Time Series with $I(0)$ and $I(1)$ Segments

Kim²⁸ showed that standard URTs are not consistent against processes displaying a shift from stationarity ($I(0)$) to nonstationarity ($I(1)$) and vice versa. Therefore new methods are required for differentiating between processes that are $I(0)$ or $I(1)$ for the entire period v.s. those with a shift from $I(0)$ to $I(1)$ or vice versa.

Kejriwal et al.²⁹ proposed hybrid testing procedures that allows ruling out of stable stationary processes or ones that are subject to only stationary changes under the null, helping the researcher in interpreting a rejection as emanating from a switch between an $I(1)$ and an $I(0)$ regime. The authors use a combination of their own developed URTs together with a test with a null of no breaks versus an alternative with one or more breaks given by Bai and Perron.²⁶ The calculation of the test statistics and the asymptotic critical values are done using the dynamic programming algorithm proposed in Perron and Qu.³⁰

Kejriwal et al.²⁹ used the approach of detrending the data when trends are included prior to using their hybrid testing procedure and they suggest using a sequential procedure developed by Kejriwal et al.²⁹ that is robust to whether the errors are $I(1)$ or $I(0)$. The smallest time series length the authors consider is 150.

Herranz²² proposed a testing procedure referred to as the Hybrid Bai-Perron-ADF procedure whereby given a time series to consider if the series follows one of the following four possibilities:

- The entire series is stationary ($I(0)$)
- The entire series is nonstationary ($I(1)$)
- The series consists of a stationary segment ($I(0)$) followed by a nonstationary segment ($I(1)$)
- The series consists of a nonstationary segment ($I(1)$) followed by a stationary segment ($I(0)$)

The proposed approach is to use the Bai and Perron²⁶ methodology of estimating structural break date/s based on finding the model specification that minimizes the RSS via a dynamic programming approach, and then to use the ADF URT to test each section to determine if it is likely $I(0)$ or $I(1)$. To do this we use the following regression Model (56) with the Bai and Perron²⁶ procedure which will estimate structural breaks in the coefficients $\beta_0, \beta_1, \beta_2$:

$$x(t) = \beta_0 + \beta_1 t + \beta_2 x(t-1) + \varepsilon_t. \quad (56)$$

Listing 3

ADF URTs on IBM Segments

```

> adf.test(IBM.$IBM.Close[1:bp1$breakpoints[1]])
Dickey-Fuller = -1.2919, Lag order = 5, p-value = 0.8717
alternative hypothesis: stationary
> adf.test(IBM.$IBM.Close[bp1$breakpoints[1]:bp1$breakpoints[2]])
Dickey-Fuller = -1.7957, Lag order = 5, p-value = 0.6615
alternative hypothesis: stationary
> adf.test(IBM.$IBM.Close[bp1$breakpoints[2]:1])
Dickey-Fuller = -2.4495, Lag order = 5, p-value = 0.3874
alternative hypothesis: stationary

```

If the Bai and Perron²⁶ structural break methodology does not find a break date, then we run the ADF test on the entire time series—and both segments would have the same result.

If we assume that if any structural changes in linear trend and/or intercept happen they do so at the same time as a structural break in AR(1) autoregressive multiplier ϕ_1 then the ADF and the Bai and Perron²⁶ methodology were highly accurate, then the ADF test would fit the correct linear trend and intercept in each segment; in any case experiments show that the Bai and Perron²⁶ approach can handle changes in β_2 across both $I(0)$ and $I(1)$ regimes, as well as when there are changes in the other coefficients.

Figure 1 shows in red vertical bars the estimated break-points using this procedure with Model (56) on historical IBM prices. When the ADF URT was run on each of the three segments, the null hypothesis of a unit root was never rejected.

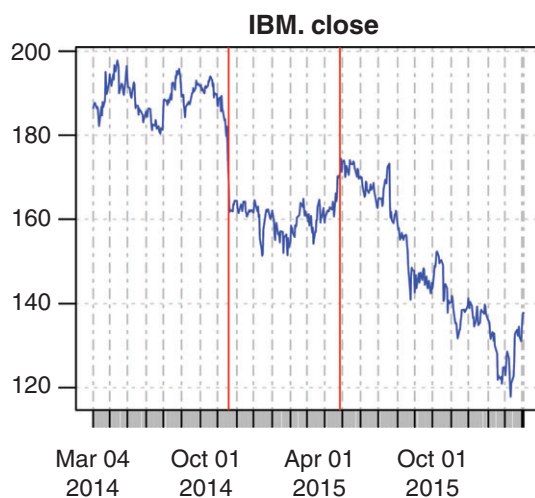


FIGURE 1 | Estimated changepoints in IBM prices.

We see in Listing 3 that the null hypothesis of a unit root cannot be rejected for any of the three segments, which leads us to conclude they are likely nonstationary: $I(1), I(1), I(1)$. This finding is consistent with the common assumption that asset prices follow a random walk. Even though breakpoints are estimated using the Bai and Perron³¹ procedure the ADF tests show that it is likely the AR(1) multiplier ϕ_1 is 1, even if the drift has a changepoint. Visually we see that there is a level-shift between the first and second segments, and there is likely a linear trend starting in the third component.

URTS ROBUST TO STRUCTURAL CHANGES UNDER THE NULL

Perron (1989) URT

Perron³² proposed URTs for processes with a linear trend with a break in intercept with the unit root as a null hypothesis, and the break model without the unit root as an alternative hypothesis:

$$\begin{aligned}
 H_0 : y_t &= y_{t-1} + \mu_1 + (\mu_2 - \mu_1)I\{t > T_B\} + \beta t + e_t, \\
 H_1 : y_t &= \mu_1 + (\mu_2 - \mu_1)I\{t > T_B\} + \beta t + e_t,
 \end{aligned} \quad (57)$$

where the innovations e_t are stationary ARMA processes:

$$A(L)e_t = B(L)v_t; v_t \text{ i.i.d}(0, \sigma^2), \quad (58)$$

with $A(L)$ and $B(L)$ p^{th} and q^{th} order polynomials, so the innovation series $\{e_t\}$ is taken to be an ARMA(p, q) process.

And the AR(1) multiplier of y_{t-1} is estimated using OLS ($\hat{\phi}_1$). As presented by Ref 10, p. 60 $\hat{\phi}_1$ is asymptotically biased towards 1, but is consistent for

Model (57), which makes this test have low statistical power.

Perron³² also considers tests for a process with breaks in intercept in trend and intercept:

$$\begin{aligned} H_0 : y_t &= y_{t-1} + \mu_1 + (\mu_2 - \mu_1)I\{t > T_B\} \\ &+ (t - T_B)(\beta_2 - \beta_1)I\{t > T_B\} + \beta_1 t + e_t, \\ H_1 : y_t &= \mu_1 + (\mu_2 - \mu_1)I\{t > T_B\} \\ &+ (t - T_B)(\beta_2 - \beta_1)I\{t > T_B\} + \beta_1 t + e_t. \end{aligned} \quad (59)$$

Choi¹⁰, p. 60 explains that $\hat{\phi}_1 \xrightarrow{p} 1$ for Model (58) implying that the URTs for this model are inconsistent: a stationary process with a breaking trend **cannot** be distinguished from a unit root process with drift.

Lee–Strazicich URTs

The one-break Lee and Strazicich³³ URT procedure and the two-break Lee and Strazicich³⁴ URT allow for structural breaks to be determined endogenously from the data and breaks are allowed under both the null and the alternative hypothesis. Consider the single break model:

$$y_t = \delta Z_t, X_t = \beta X_{t-1} + \varepsilon_t, \quad (60)$$

where Z_t contains exogenous variables. The null hypothesis is specified by $\beta = 1$.

‘Model A’ is the crash model that allows for a single change intercept where $Z_t = [1, t, D_t]'$ where $D_t = 1$ for $t \geq T_B + 1$, and zero otherwise. T_B is the time of the structural break and $\delta = (\delta_1, \delta_2, \delta_3)$

‘Model C’ allows for a shift in intercept and change in trend slope under the alternative hypothesis where $Z_t = [1, t, D_t, DT_t]'$ where $DT_t = t - T_B$ for $t \geq T_B + 1$, and zero otherwise. T_B is the time of the structural break and $\delta = (\delta_1, \delta_2, \delta_3)$.

The URT statistics are derived from the regression:

$$\Delta y_t = \delta' \Delta Z_t + \phi \tilde{S}_{t-1} + u_t, \quad (61)$$

$$\tilde{S}_t = y_t - \Psi_x - Z_t \tilde{\delta}, t = 2, \dots, T, \quad (62)$$

where $\tilde{\delta}$ is the coefficients estimated in the regression of Δy_t on ΔZ_t and $\tilde{\Psi}_x$ is the restricted MLE of $\Psi_x = \Psi + X_0$ given by $y_1 - Z_1 \tilde{\delta}$. The unit root null hypothesis consists of $\phi = 0$ and the LM(Lagrange multiplier) t -test statistic $\tilde{\tau} = t$ -statistic testing the null hypothesis $\phi = 0$. As in the case with the ADF test, a correction

for auto-correlated innovations is made by adding lagged terms $\Delta \tilde{S}_{t-j}$ where $j = 1, \dots, k$. The location of the break T_B is determined by searching across all breakpoints and picking the one with the most negative $\tilde{\tau}$:

$$\inf \tilde{\tau}(\tilde{\lambda}) = \inf_{\lambda} \tilde{\tau}(\lambda) \text{ where } \lambda = T_B/T. \quad (63)$$

If the DGP is that in (60), and if the ε_t satisfy certain regularity conditions, and $T_B/T \rightarrow \lambda$ as $T \rightarrow \infty$, then under the null hypothesis of $\beta = 1$

$$\inf \tilde{\tau}(\tilde{\lambda}) = \inf_{\lambda} \left(-\frac{1}{2} \int_0^1 \underline{V}(r)^2 dr \right)^{-1/2}, \quad (64)$$

where $\underline{V}(r)$ represents a demeaned Brownian bridge.

Hybrid Bai-Perron-Zivot-Andrews URT

Herranz²² proposed a new URT that allows structural breaks under the null hypothesis, which we refer to here as the Hybrid Bai-Perron-Zivot-Andrews (HBPZA) URT, and consists of the following steps:

1. Given a time series, first use the Bai-Perron break-point estimation procedure in Ref 31 using regression Model (65) to detect changes in the coefficients. This divides the time series into $k + 1$ segments given a total of k breakpoints.
2. For each segment within the time series compute the Zivot Andrews (ZA) URT statistic.
3. A final test statistic is computed by weighing each sub-test statistic by the segment length.
4. If there are more than three breakpoints estimated, only the first three breakpoints are used to determine four segments.

The regression Model (65) is used to detect breakpoints in the coefficients β_0 , β_1 , and β_2 using the Bai and Perron³¹ procedure:

$$x_t = \beta_0 + \beta_1 t + \beta_2 x_{t-1} + \varepsilon_t. \quad (65)$$

DGP (65) was used to simulate the time series under the null hypothesis (H_0) of a unit root with 1 structural break at time T_B to derive the critical values of the test statistic (via simulation):

$$\begin{aligned} y_t &= \phi_1 y_{t-1} + \mu_1 + (\mu_2 - \mu_1)I\{t > T_B\} \\ &+ (t - T_B)(\beta_2 - \beta_1)I\{t > T_B\} + \beta_1 t + e_t, \end{aligned}$$

TABLE 3 | Hybrid Bai-Perron-Zivot-Andrews Unit Root Critical Values from Simulations

Breaks	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.1$
	-6.069573	-4.631409	-4.336683
	-4.837634	-4.376655	-4.178676

$$\begin{aligned} \mu_1 &\sim U(-10, 10); \mu_2 \sim U(-10, 10); y_0 \sim N(0, 1) \\ \beta_1 &\sim U(-3, 3); \beta_2 \sim U(-3, 3), \\ H_0 : \phi_1 &= 1; H_A : \phi_1 \neq 1. \end{aligned} \quad (66)$$

This approach can be extended to two or more breakpoints. The HBPZA critical values for 0 and 1 breakpoints under the null hypothesis of a unit root derived from simulations are detailed in Table 3.

URTS AND MORE AVAILABLE IN THE R LANGUAGE

Table 4 details URTs available as functions in R programming language packages; most of them were used in simulations detailed in the next section. Note that the URTs in the fUnitRoots package³⁵ are the same as those available in the urca package.²⁰

The Lee–Strazicich (LS) URT function `ls()` used in this article was written by Raphael Rocha Gouvea and Thais Helena Fernandes Teixeira and was obtained from the open knowledge repository of the Institute for Applied Economic Research (Ipea)—a federal public foundation linked to the Secretariat of Strategic Affairs of the Presidency of the Republic of Brazil.^a

The `auto.arima()` function of the R forecast package can be used to fit the optimal ARIMA() model to data.

Three-regime SETAR models of time series can be defined and fitted using the `setar()` function of the `tsDyn` R package.³⁶ For known thresholds the SETAR model can be fit using conditional least squares; the `setar()` function performs a grid search to find the thresholds that minimize the RMSE.

The Johansen cointegration test used in this article is the `ca.jo()` function of the `urca` R package by Ref 20. The Bai and Perron²⁶ methodology to determine break dates in time series used in this article is the `breakpoints()` function of the R `strucchange` package.

Box 1

UNIT ROOTS IN VARIOUS FIELDS

As previously mentioned in the introduction, many financial time series such as asset prices are modeled as unit root processes. If you pick any time series of stock prices, or interest rates, under normal conditions, and run a URT on them you will likely not be able to reject the null hypothesis of a unit root.

The bulk of the literature on unit roots and cointegration spans mostly the social sciences, largely focusing on economic and financial applications. However, there are other fields where they may be proving to be useful tools.

Kipinski et al.³⁷ used Phillips-Perron URTs as well as other statistical tests to investigate the stationarity of magnetoencephalography (MEG) or electroencephalography (EEG) time series, and they found they are largely stationary. However, Koruek and Ozkaya³⁸ analyzed EEG time series before, during and after seizures, and they determined that short interictal (between seizures) EEG series were nonstationary and could be modeled as an ARIMA unit root process.

There have been various studies on testing worldwide zonal temperature anomalies for unit roots under various assumptions of structural changes as in Refs 39–41. A three regime SETAR model of the Northern Hemisphere temperature anomalies can provide a better model that incorporating changepoints than breaks in an AR framework. This is the case described in Whyte and Metcalfe⁴² where they propose a model similar to SETAR which they refer to as MTAR using a reconstructed Burgundy temperature series of Chuine et al.⁴³ from 1370 to 1977. The constants are negative but the linear trends are positive indicating a diverging relationship further exacerbated by an nonstationary(explosive) middle regime.

Sobradelo et al.⁴⁴ used a Dickey-Fuller URT to provide evidence that a time series of volcanic eruptions is not stationary. Ramana et al.⁴⁵ used ADF URTs to show that water levels in reservoirs are likely nonstationary, prior to testing for cointegration.

Vargas and Kastle⁴⁶ assumed that geological magnetic time series are unit roots to consider a cointegration argument. Oluwole⁴⁷ used a URT to determine the likelihood that a time series of outbreaks of meningitis was stationary. Skrastins et al.⁴⁸ used ADF URTs to check for the stationarity of web traffic time series.

TABLE 4 | Unit Root Tests in R

Package	Function Name	Unit Root Test	Constant	Linear Trend	Constant and Linear Trend	Driftless
tseries	adf.test	Augmented Dickey Fuller	N.A.	N.A.	Only choice	N.A.
fUnitRoots	urdfTest	Augmented Dickey Fuller	'c'	N.A.	'ct'	'nc'
fUnitRoots	urersTest	Elliott Rothenberg Stock	'c'	N.A.	'ct'	N.A.
fUnitRoots	urkpsTest	KPSS for stationarity	'mu'	N.A.	'tau'	N.A.
fUnitRoots	urppTest	Phillips-Perron	'constant'	'trend'	N.A.	N.A.
fUnitRoots	urspTest	Schmidt Phillips	N.A.	N.A.	N.A.	N.A.
fUnitRoots	urzaTest	Zivot Andrews	'intercept'	'trend'	'both'	N.A.
tsDyn	BBCTest	BBC unit root test against a stationary 3 regime SETAR alternative	Only choice	N.A.	N.A.	N.A.

URT SIMULATIONS

URTs on AR(1) with Intercept and a Linear Trend

Table 5 summarizes the results of simulations performed with DGP (67) with initial value $x_0 = 20$. This process consists of AR(1) processes with a deterministic intercept and a linear trend with independent

identically distributed Gaussian innovations for various values of ϕ_1 .

The results can be summarized as follows:

- Data simulated under the alternative hypothesis where $\phi_1 > 1$: The ERS, ADF and ZA tests do not significantly reject the null hypothesis of even though ideally they should in this case. However the two lagged-series based tests reject 97% of the time for $\phi_1 = 1.04$.
- Data simulated under the null hypothesis: Hypothesis where $\phi_1 = 1$: The URTs reject the null hypothesis as expected close to the significance level of 5%: ADF at 4%, and lagged-series and ERS-DFGLS, ZA tests at 5% and the ERS-Ptest is invalid.
- Data simulated under the alternative hypothesis where $\phi_1 < 1$: The null hypothesis is rejected at a rate of 10% or less as follows from best to worst:
 - ADF and lagged-series, ZA and ZA-lagged-series tests: $\phi_1 \leq 0.5$
 - The ERS tests hardly ever reject the null hypothesis in this case. In other tests when the initial value of $x_0 = 0$ the statistical power is similar to that of the ADF URT.

TABLE 5 | Proportions of Failures to Reject the Unit Root Null Hypothesis of URTs on AR(1) Processes with Intercept $U(-1,1)$ and Linear Trend $U(-1,1)$ with $l = 100$, $m = 5000$, and $\alpha = 5\%$ and $x_0 = 20$

ϕ_1	ADF	Lagged-series	ERS-Ptest	ERS-DFGLS	ZA-lagged-series	ZA
1.1000	1.00	0.03	1.00	1.00	0.21	1.00
1.0800	1.00	0.02	1.00	1.00	0.18	1.00
1.0600	1.00	0.02	1.00	0.93	0.18	1.00
1.0400	1.00	0.03	0.88	0.98	0.20	1.00
1.0200	0.99	0.74	0.80	0.99	0.91	0.97
1.0100	0.96	0.94	0.36	0.96	0.96	0.94
1.0050	0.96	0.95	0.34	0.95	0.96	0.94
1.0025	0.96	0.95	0.35	0.95	0.95	0.94
1.0000	0.96	0.95	0.34	0.95	0.96	0.95
0.9950	0.96	0.95	0.34	0.95	0.95	0.95
0.9800	0.95	0.94	0.34	0.95	0.95	0.94
0.9600	0.94	0.92	0.37	0.96	0.94	0.93
0.9000	0.81	0.75	0.48	1.00	0.81	0.87
0.7000	0.24	0.18	0.56	1.00	0.24	0.13
0.5000	0.08	0.02	0.58	1.00	0.07	0.00
0.3000	0.04	0.00	0.60	1.00	0.01	0.00
0.1000	0.03	0.00	0.61	1.00	0.00	0.00
0.0000	0.02	0.00	0.61	1.00	0.00	0.00

$$\begin{aligned}
 y_t &= \phi_1 x_t + \alpha + \beta t, \\
 x_t &= x_{t-1} + \varepsilon_t, \\
 x_0 &= 0; \varepsilon_t \sim i.i.d. \sim N(0,1), \\
 t &= 1, \dots, l, \\
 \alpha &\sim U(-1,1), \beta \sim U(-1,1).
 \end{aligned} \tag{66}$$

Table 6 shows simulations of Model (67) using negative values of ϕ_1 . The results for data simulated under the alternative hypothesis where $\phi_1 < 0$ can be

TABLE 6 | Proportions of Failures to Reject the Null Hypothesis of a Unit Root of URTs on Driftless AR(1) Processes with $\phi_1 < 0$ and $l = 1000$, $m = 1000$, and $\alpha = 5\%$ and $x_0 = 20$

ϕ_1	ADF	Lagged-series	ERS-Ptest	ERS-DFGLS	ZA-lagged-series	ZA
-1.0000	0.01	0.00	0.99	0.99	0.00	0.00
-0.9900	0.01	0.00	0.98	0.97	0.00	0.00
-0.9500	0.01	0.00	0.87	0.97	0.00	0.00
-0.9000	0.01	0.00	0.76	0.98	0.00	0.00
-0.7000	0.01	0.00	0.66	0.99	0.00	0.00
-0.5000	0.01	0.00	0.63	1.00	0.00	0.00
-0.3000	0.01	0.00	0.62	1.00	0.00	0.00
-0.1000	0.02	0.00	0.61	1.00	0.00	0.00
0.0000	0.02	0.00	0.61	1.00	0.00	0.00
1.0000	0.96	0.95	0.34	0.95	0.96	0.95

summarized as follows: The null hypothesis is rejected as follows from best to worst:

- lagged-series, ZA and ZA-lagged-series fully reject the null and the ADF test almost fully for all tested negative values of ϕ_1 clearly outperforming the ERS tests, which perform quite poorly in these cases with negative ϕ_1 .

URT on AR(1) with Breaks

Table 7 summarizes the results of simulations performed with DGP (68). This model consists of AR(1)

TABLE 7 | Proportions of Failures to Reject Unit Root Null Hypothesis of URTs on AR(1) Processes with one Structural Break in Intercept $U(-1,1)$ and Linear Trend $U(-1,1)$ with $l = 200$, $m = 1000$, $x_0 \sim N(0, 1)$ and $\alpha = 5\%$

ϕ_1	ADF	Lagged-series	ERS-Ptest	ERS-DFGLS	ZA-lagged-series	ZA
1.010	0.98	0.96	0.98	0.98	0.25	0.25
1.005	0.97	0.98	0.98	0.98	0.24	0.24
1.000	0.97	0.98	0.98	0.98	0.23	0.23
0.990	0.97	0.97	0.98	0.98	0.23	0.23
0.980	0.97	0.97	0.97	0.97	0.23	0.23
0.970	0.96	0.96	0.96	0.96	0.21	0.21
0.960	0.95	0.96	0.94	0.95	0.20	0.20
0.950	0.95	0.95	0.93	0.94	0.18	0.18
0.900	0.92	0.92	0.87	0.88	0.12	0.13
0.000	0.88	0.85	0.93	0.88	0.00	0.00

processes with one structural break in intercept and one structural break in linear trend with independent identically distributed Gaussian innovations for various values of ϕ_1 . The break time is determined at random; it is uniformly distributed, and the same break time is used for the intercept change and the linear trend change. It is tested with two different sets of intercept and trend levels.

$$\begin{aligned}
 y_t &= x_t + \alpha_1 + (\alpha_2 - \alpha_1)I\{t > t_u\} + \beta_1 t \\
 &\quad + (\beta_2 - \beta_1)(t - t_u)I\{t > t_u\}, \\
 x_t &= \phi_1 x_{t-1} + \varepsilon_t; \varepsilon_t \sim i.i.d. \sim N(0,1), \\
 x_0 &\sim N(0,1), \\
 t_u &\sim U(3, l-2); t = 1, \dots, l.
 \end{aligned} \tag{68}$$

The results can be summarized as follows. For DGP (68) the intercepts and trends are picked uniformly random for each simulation as follows: $\alpha_1, \alpha_2, \beta_1, \beta_2 \sim U(-1,1)$:

- Data simulated under the alternative hypothesis where $\phi_1 > 1$: The ERS, ADF and ZA tests do not significantly reject the null hypothesis of a unit root even though ideally they should in this case. However the ZA and the ZA-lagged-series tests reject 75% of the test cases when $\phi_1 = 1.01$.
- Data simulated under the null hypothesis where $\phi_1 = 1$: Four of the URTs reject the null hypothesis as expected close to the significance level of 5%: ADF 3%, ERS-Ptest 2%, lagged-series 2% and ERS-DFGLS 2%. However the ZA and the ZA-lagged-series tests have a rejection rate of 77% under the null. The ZA related tests are the worst performing test under the null hypothesis.
- Data simulated under the alternative hypothesis where $\phi_1 < 1$: The null hypothesis is never significantly rejected with the ADF, the lagged-series and both ERS tests regardless of the value of ϕ_1 . The ZA and the ZA-lagged-series tests reject the null hypothesis 100% or more of the cases when $\phi_1 \leq 0.0$.

The results are understandable as the Zivot Andrews URT does not allow breaks under the null; it only allows breaks under the alternative hypothesis. When using standard URTs such as the ADF test it is not possible to distinguish a stationary process with structural breaks in the linear trend from a unit root with a drift(trend) as detailed in Ref 10, p. 60.

LS vs Hybrid Bai-Perron-Zivot-Andrews

The LS URT³⁴ and the Hybrid Bai-Perron Zivot-Andrews URT proposed in Herranz²² both allow structural breaks of the intercept and/or linear trend under the null hypothesis of a unit root. Simulations were performed with AR(1) time series with one structural break using Model (68) with $\alpha_1 = 50$, $\alpha_2 = 1000$, $\beta_1 = 1$, $\beta_2 = 3$ under various levels of ϕ_1 . The break times were 25% and 50% of the time series length. Two sets of tests were performed, one for each break ratio, with a time series of length 100 and are detailed in Table 8.

The summary of results is as follows:

- Data simulated under the alternative hypothesis where $\phi_1 = 1.01$: The HBPZA test, nor the LS test significantly reject the null hypothesis of a unit root even though ideally they should in this case.

TABLE 8 | Proportions of Failures to Reject the Unit Root Null Hypothesis of URTs on AR(1) Processes with 1 Break in Intercept and Trend with $l = 100$, $m = 1000$, and $\alpha = 5\%$, $\alpha_1 = 50$, $\alpha_2 = 1000$, $\beta_1 = 1$ and $\beta_2 = 3$

ϕ_1	Bai-Perron-Zivot-Andrews	Lee-Strazicich	Breakpoint Proportion
1.0100	0.97	0.99	0.25
1.0000	0.97	0.99	0.25
0.9700	0.96	0.99	0.25
0.9500	0.96	0.98	0.25
0.9000	0.96	0.97	0.25
0.7000	0.87	0.65	0.25
0.5000	0.53	0.20	0.25
0.4000	0.28	0.18	0.25
0.2500	0.05	0.21	0.25
0.1000	0.00	0.25	0.25
0.0000	0.00	0.27	0.25
1.0100	0.96	0.99	0.50
1.0000	0.96	0.99	0.50
0.9700	0.96	0.99	0.50
0.9500	0.96	0.99	0.50
0.9000	0.96	0.97	0.50
0.7000	0.83	0.62	0.50
0.5000	0.37	0.22	0.50
0.4000	0.18	0.18	0.50
0.2500	0.04	0.22	0.50
0.1000	0.01	0.26	0.50
0.0000	0.00	0.29	0.50

- Data simulated under the null hypothesis where $\phi_1 = 1$: The HBPZA and LS tests reject the null hypothesis less than the significance level of 5% for both break times of 50 and 25.
- Data simulated under the alternative hypothesis where $\phi_1 < 1$: When the break time is 25 and 50 the HBPZA test significantly outperforms the Lee-Strazicich test in this scenario when $\phi_1 \leq 0.25$.

Threshold URTs

The three regime SETAR DGP (68) was used for simulations under various combinations of the autoregressive multipliers ϕ_1^H , ϕ_1^M , and ϕ_1^L corresponding to the high(H), middle(M) and low(L) regimes. Two different sets of symmetric thresholds were used ($\theta_H = 2$, $\theta_L = -2$) and ($\theta_H = 20$, $\theta_L = -20$). Threshold models are complex to fit as there are many factors that affect the quality of the fit including the amount of data in each regime, and the number of parameters that need to be fit, and the length of the time series. If the threshold parameters are known, then conditional least squares can be used to fit the model.

$$z_t = \begin{cases} \phi_1^H z_{t-1} + \varepsilon_t, & \text{if } z_{t-1} > \theta_H, \\ \phi_1^M z_{t-1} + \varepsilon_t, & \text{if } \theta_L \leq z_{t-1} \leq \theta_H, \\ \phi_1^L z_{t-1} + \varepsilon_t, & \text{if } z_{t-1} < \theta_L, \end{cases} \quad (69)$$

$$z_0 \sim N(0,1).$$

Stigler⁶ points out that DGP (69) is stationary when $\phi_1^H < 1$ and $\phi_1^L < 1$. The AR multiplier ϕ_1^M in the middle regime does not matter much as long as the high and/or low regimes are being activated. If the realization of the process were to always stay in the middle regime because the threshold values were never reached, then for all practical purposes this regime would be the important one to consider. Simulations were performed under different combinations of AR multipliers for each high(ϕ_1^H), middle(ϕ_1^M), and low(ϕ_1^L) regimes, for two sets of symmetric thresholds ($\theta_L = -2$, $\theta_H = 2$) and ($\theta_L = -20$, $\theta_H = 20$) and tests were performed for a time series of length 200.

The results for tests with time series of length 200 are in Table 9 and can be summarized as follows:

- Trivial unit root($\phi_1^H = \phi_1^M = \phi_1^L = 1$): As expected the tests reject close to the 5% rate of test

TABLE 9 | Proportions of Failures to Reject the Unit Root Null Hypothesis of Unit Root Tests on Various Threshold AR(1) Processes with $l = 200$, $m = 1000$, and $\alpha = 0.05$, $z_0 = 0$

θ^H	θ^L	ϕ_1^H	ϕ_1^M	ϕ_1^L	ADF	Lagged-series	BBC
2.0	-2.0	1.00	1.00	1.00	0.94	0.96	0.95
2.0	-2.0	0.00	0.00	0.00	0.00	0.00	0.00
2.0	-2.0	1.00	0.90	1.00	0.90	0.95	0.95
2.0	-2.0	1.00	1.10	1.00	0.96	0.95	0.95
2.0	-2.0	1.01	1.10	1.01	0.99	0.96	0.98
2.0	-2.0	0.90	1.00	0.90	0.01	0.44	0.80
2.0	-2.0	0.90	1.10	0.90	0.02	0.57	0.70
20.0	-20.0	1.00	0.90	1.00	0.01	0.32	0.83
20.0	-20.0	1.00	1.10	1.00	1.00	0.44	0.54
20.0	-20.0	1.01	1.10	1.01	1.00	0.95	0.81
20.0	-20.0	0.90	1.00	0.90	0.94	0.95	0.89
20.0	-20.0	0.90	1.10	0.90	1.00	0.20	0.00

cases as dictated by the significance level of 0.05: ADF 6%, lagged-series 4% and BBC 5%.

- Unit root ($\phi_1^H = \phi_1^L = 1, \phi_1^M < 1$): When the threshold range is $(\theta_L = -2, \theta_H = 2)$ all the tests perform relatively well and reject only close to 5% of the test cases as expected. When the threshold range is $(\theta_L = -20, \theta_H = 20)$ with $\phi_1^M = 0.9$ the results are significantly worse where the lagged-series tests reject 68% of the tests of a unit root and the ADF test rejects 99% of the cases. However the BBC only rejects 17% of the cases; the process spends considerable more time in the middle regime than with the smaller thresholds. When $\phi_1^M = 1.1$ the ADF tests do not reject the null root hypothesis at all, and the lagged-series and the BBC test reject much more, 56% and 46% respectively.
- Stationary process ($\phi_1^H = \phi_1^L = 0.9$): When the threshold range is $(\theta_L = -2, \theta_H = 2)$ only the ADF test sufficiently rejects the null hypothesis of a unit root. When the threshold range is $(\theta_L = -20, \theta_H = 20)$ when $\phi_1^M = 1$ all of the tests do not significantly reject the null of a unit root. When the middle regime is explosive ($\phi_1^M = 1.1$) the BBC tests fully reject all cases of a unit root, the lagged-series rejects 80% of the cases and the ADF does not reject any.

TABLE 10 | Proportions of Failures to Reject the Unit Root Null Hypothesis on a AR(1) Series with a Break in ϕ_1 , $l = 100$, $m = 1000$ and $\alpha = 0.05$

Break Proportion	ϕ_1^A	ϕ_1^B	ADF ^A	ADF ^B	ADF ^{AB}
0.50	0.000	0.000	0.06	0.06	0.06
0.50	1.000	1.000	0.94	0.94	0.96
0.50	1.000	0.500	0.93	0.65	0.91
0.25	0.000	0.000	0.06	0.06	0.06
0.25	1.000	1.000	0.94	0.94	0.96
0.25	1.000	0.500	0.79	0.43	0.74

- Fully explosive process ($\phi_1^H = 1.01, \phi_1^M = 1.1, \phi_1^L = 1.01$): All tests hardly reject the null hypothesis of a unit root.

For the previously discussed SETAR simulations not one of the examined URTs outperformed all of the others in every case. It is clear that the time series length is a significant factor in these tests, as well as of course the threshold settings. If the process does not have sufficient data in all thresholds then it is impossible to expect the tests to work across all thresholds.

Simulations with the Hybrid Bai-Perron ADF $I(0)/I(1)$ Testing Procedure

$$y_t = \phi_1^A y_{t-1} I\{t \leq T_B\} + \phi_1^B y_{t-1} I\{t > T_B\}, \quad (70)$$

$$y_0 \sim N(0, 1).$$

Simulations performed using the DGP (69) which has breaks only in the AR ϕ_1 multiplier with a time series of length $l = 100$ are summarized in Table 10.

We can see that this new Hybrid Bai-Perron-ADF testing procedure is sensitive to the time series length and the location of the structural break. When compared to using a single URT on the entire time series, this approach can be significantly more accurate, as can be seen in the case with $\phi_1^A = 1$ and $\phi_1^B = 0.5$ where we make a greater error if we assume that the entire series is homogeneous $I(1)$ or $I(0)$.

CONCLUSIONS

Unit roots can lead to spurious regressions if used in linear regression analyses so we must test our time series with URTs beforehand to ensure they are likely stationary processes. If with each time series tested

we cannot reject the null hypothesis of a unit root, then a cointegration analysis should be pursued, however we must be aware that near-unit root processes can result in spurious cointegration. There is no single URT that is optimal under every scenario. There are complications such as considering whether or not to include deterministic components such as linear trends or not under the null hypothesis, or how long of a lag we should use in a URT. In the case where there are changepoints/breaks in the deterministic components under the null hypothesis we must be aware that most URTs, such as the ADF test,¹⁸ will not have sufficient statistical power to ever reject the null hypothesis. In these cases a URT such as the LS⁴⁹ should be used.

We must also realize that most of the URTs assume that the time series tested do not exhibit changepoints in the level of the auto-regressive multiplier ϕ_1 . Consider testing the underlying time series for breaks in all relevant parameters, including ϕ_1 using

the Bai and Perron³¹ structural break detection methodology, and then testing each time series segment with a URT such as ADF.¹⁸ In this paper we only considered $I(0)$ and $I(1)$ time series, and did not discuss $I(2)$ or higher integration orders of time series, nor did we consider fractionally integrated time series.^b

NOTES

^a <http://repositorio.ipea.gov.br/>

^b In this article it was not possible to review entirely the ever increasing literature on URTs and related theory. A good introduction providing some of the tools necessary to derive asymptotic formulas in URTs is Ref 14. A relatively short yet very comprehensive book on URTs which has been referred to in this article is Ref 10, which includes many topics not included in this review such as additional URTs, fractional unit roots, panel unit roots, seasonal unit roots and Bayesian inference methods in unit roots. Perhaps the most extensive and complete books on URTs covering key concepts and extensions Refs 11 and 12.

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